

MOHAMMAD RAHMANITALAB

B.Sc. Mechanical Engineering (expected 2026), Sharif University of Technology

Minor Computer Science (expected 2026), Sharif University of Technology

📞 (+98) 939-0053-773 | ✉️ mohammad.rahmanitalab@sharif.edu | in Mohammad-Rahmanitalab | 📍 Tehran, Iran

EDUCATION

B.Sc. Major in Mechanical Engineering, SUT | Tehran, Iran 2021 – 2026

Advisor: Prof. Saeed Behzadipour

- GPA: 3.95/4.00 (18.93/20) — Ranked **12th out of 116** students in the department.
- Qualified for **direct M.Sc. admission** based on outstanding academic excellence.

B.Sc. Minor in Computer Science, SUT | Tehran, Iran 2022 – 2026

- GPA: 4.00/4.00 (18.93/20)

Diploma in Mathematics and Physics | Tehran, Iran 2021

- GPA: 4.00/4.00 (19.88/20)

RESEARCH INTERESTS

- **Medical Robotics:** Surgical robots, exoskeletons, rehabilitation robotics, human–robot interaction
- **Motion Analysis:** Biomechanics, IMU-based sensing, clinical movement classification
- **Machine Learning:** Deep learning, reinforcement learning, ML for healthcare applications
- **Control Systems:** Haptic feedback, teleoperation, master–slave robotic control

UNDERGRADUATE RESEARCH

Motion Analysis for DMD Rehab | Sharif University of Technology 2024 – Present

- Conducting motion analysis using IMU sensors to develop evidence-based therapeutic protocols for children with Duchenne Muscular Dystrophy (DMD), supervised by Prof. Behzadipour.
- Leading the full pipeline: data collection, preprocessing, feature extraction, and movement classification using machine learning and deep learning models (MATLAB, Python).
- Investigating movement biomarkers to support clinical decision-making in pediatric rehabilitation.

PROFESSIONAL EXPERIENCE

Back-End Developer | Moj Gostaran Adak, Tehran Oct 2025 – Dec 2025

- Developed a Java-based backend server integrated with the Traccar GPS tracking platform, enabling reliable real-time device communication and data handling.
- Designed and implemented a customized Go backend for the company's internal frontend, supporting a locker-tracking system deployed in the Hamrah Aval project.
- Built optimized RESTful APIs ensuring seamless interoperability between tracking services and client applications.

INTERNSHIPS

Robotics Engineering | Sina Robotics & Medical Innovators Co., Tehran Jun – Sep 2024

- Worked on surgical robotics with emphasis on kinematics; derived analytical solutions for master–slave robotic systems to improve precision and responsiveness (MATLAB, MSC ADAMS, TwinCAT 3).
- Contributed to the design, simulation, and testing of robotic components, with focus on stability and haptic feedback in teleoperated surgical systems.

TEACHING ASSISTANT

Fundamental Automatic Control Design | Sharif University of Technology Fall 2025

- Assisted in instruction of controller design, PID tuning methods, and stability analysis.
- Managed grading of assignments, quizzes, and exams; held office hours for student guidance.

Automatic Control | Sharif University of Technology Spring 2025

- Supported lectures on control theory, feedback systems, and Lyapunov stability analysis.
- Assisted with course assessment and provided academic mentoring to students.

AWARDS AND HONORS

- Ranked 85th nationally in the National Master's Entrance Exam (Konkur), Mechanical Engineering. 2025
- Qualified for direct admission to M.Sc. programs ("without exam") due to academic excellence. 2025 & 2026
- Ranked 11th (top 11%) among 116 students in the Mechanical Engineering Department. Present
- Ranked 187th nationally in the National University Entrance Exam (Konkur), among ~500,000 applicants. 2021

COURSE PROJECTS

Robotics and Control

Robotics

Fall 2024

Modeled and simulated a multi-DOF robotic manipulator in MATLAB/Simulink, studying forward/inverse kinematics and dynamic behavior.

Fundamental Automatic Control Design

Fall 2024

Designed and tuned a PID controller for a ball-and-beam system; analyzed transient response and stability margins in MATLAB/Simulink.

Automatic Control

Spring 2024

Designed a PID controller for a rotating governor; performed full system performance and stability analysis.

Dynamics of Machines

Spring 2024

Simulated a pushing mechanism in MATLAB/Simulink to analyze forces, motion, and dynamic response.

Programming

Advanced Programming (Java)

Spring 2025

Developed an online multiplayer game using object-oriented design, multithreading, and socket-based network programming.

Fundamentals of Programming (C)

Spring 2022

Implemented a Twitter-like social media application with user management and file-based data persistence.

SKILLS

Languages: Persian (Native), English (Excellent).

Programming: Java (advanced), C (advanced), Go (intermediate), C++ (intermediate), Python (intermediate), TwinCAT 3 (intermediate)

Simulation: MATLAB/Simulink (advanced), ROS (beginner), Gazebo (beginner)

CAD and Dynamics: SolidWorks, MSC ADAMS

Soft Skills: Clean coding, technical writing, public speaking, team leadership

- Fundamentals of Programming (C)
- Advanced Programming (Java)
- Discrete Mathematics
- Probability
- Linear Algebra
- Operating Systems

- Dynamics of Machines
- Robotics
- Automatic Control
- Fundamental Automatic Control Design
- Data Transmission and Networks
- Computer Simulation

SELECTED COURSES

EXTRA-CURRICULAR ACTIVITIES

CARE Student Society: Served as Media Director, managing media content, production, and team operations.

ICERS 2024: Contributed to event organization and media coordination as a member of the IT and Media Team.

Dept. Website: Developed and maintained the official website of the Mechanical Engineering Department (WordPress).